A NONERGODIC FLOW RULE WITH GALTON–WATSON STATIONARY MEASURE

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Abstract. A flow rule assigns to every infinite rooted tree a unit flow from the root to infinity, consistently with passage to the subtrees of the children of the root, and it defines a Markov chain on the space of trees that moves from a tree to the subtree of a child chosen according to the flow. Lyons, Pemantle and Peres showed that the chain is ergodic when a stationary measure absolutely continuous with respect to a Galton–Watson measure GW exists and the flow is positive on every child, and asked whether a flow rule with a stationary measure equivalent to GW must give an ergodic chain. It need not. For the Galton–Watson measure with offspring distribution $p_2 = 3/5$, $p_3 = 2/5$ there is a flow rule for which GW itself is stationary and the stationary chain has an invariant event of probability $1/2$. The rule follows a colouring of the vertices that is a measurable function of the tree, in which a vertex with two children takes the AND of their colours and a vertex with three children the OR: the flow is spread evenly over the children that have the colour of the root. The two colour classes give the ergodic decomposition of the chain.

1. Introduction

Let $\mathcal{T}$ be the space of rooted, labelled, locally finite trees with its usual metric, in which two trees are close when they agree up to a large generation [4, Exercise 5.2]. For a tree T and a vertex x of T, let T^x be the subtree of descendants of x, rooted at x, and write $|x|$ for the generation of x. A *flow* on T is a unit flow from the root to infinity; we write $\theta(x)$ for the amount of flow through the vertex x, so that $\theta(\text{root}) = 1$ and $\theta(x)$ is the sum of θ over the children of x. If $\theta(x) > 0$, then $\theta^x(y) = \theta(y)/\theta(x)$, $y \in T^x$, is a flow on T^x. Since a finite tree carries no flow, these notions concern the Borel set $\mathcal{T}_\infty$ of infinite trees.

Following Lyons, Pemantle and Peres [5] (see [4, Section 17.5]), a *flow rule* is a Borel map Θ that assigns to every $T \in \mathcal{T}_\infty$ a flow Θ_T on T such that

$$|x| = 1,\ \Theta_T(x) > 0 \implies (\Theta_T)^x = \Theta_{T^x},$$

and, as Lyons and Peres assume throughout, that is equivariant: $\Theta_{\varphi(T)}(\varphi(x)) = \Theta_T(x)$ for every rooted isomorphism φ of T onto a tree $\varphi(T)$. A flow rule defines the Markov chain on $\mathcal{T}_\infty$ with transition kernel

$$p(T, \cdot) = \sum_{|x|=1} \Theta_T(x)\, \delta_{T^x}.$$

A probability measure μ is Θ-*stationary* if $\mu(A) = \int p(T, A)\, d\mu(T)$ for every Borel set A; a Borel set A is *invariant* if $p(T, A) = \mathbf{1}_A(T)$ for μ-a.e. T; and the chain with initial distribution μ is *ergodic* if every invariant set has μ-measure 0 or 1. For stationary μ this is equivalent to the ergodicity of the shift on the space of paths of the chain [4, Corollary 17.3].

Let GW be a Galton–Watson measure on $\mathcal{T}$ whose offspring distribution (p_k) satisfies $p_0 = 0$ and $p_k < 1$ for all k, the standing assumption of [4, Chapter 17]; then GW-a.e. tree is infinite.

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Proposition 17.17 of [4] (Proposition 5.2 of [5]) states that if $\Theta_T(x) > 0$ for GW-a.e. T and every $|x| = 1$, then the chain with initial distribution GW is ergodic, and a Θ-stationary measure that is absolutely continuous with respect to GW is equivalent to GW and gives an ergodic stationary chain. Rousselin [6] constructs explicit stationary measures for a class of flow rules on Galton–Watson trees whose vertices carry i.i.d. marks, and uses them to compute the dimension of harmonic measures. Lyons and Peres write that they do not know whether the first hypothesis is necessary for the final conclusion, and ask [4, Question 17.18], as in [5, Section 10, Question 3]:

> If a flow rule has a stationary measure equivalent to GW, must the associated Markov chain be ergodic?

Without the positivity hypothesis absolute continuity does not imply equivalence: in [4, Exercise 17.22], with $p_3 = p_4 = 1/2$, the flow follows an n-ray, a ray along which every vertex has exactly n children, when there is one, and the stationary measure $(\mathrm{GW}_{A_3} + \mathrm{GW}_{A_4})/2$, where A_n is the event that the tree contains an n-ray, is absolutely continuous with respect to GW and gives a nonergodic chain; it is not equivalent to GW, since $\mathrm{GW}(A_3 \cup A_4) < 1$. The answer to the question is also negative.

Theorem 1. *Let* GW *be the Galton–Watson measure with offspring distribution* $p_2 = 3/5$, $p_3 = 2/5$. *There are a flow rule* Θ *and an isomorphism-invariant Borel set* $B \subset \mathcal{T}_\infty$ *with the following properties.*

(i) GW *is* Θ*-stationary.*
(ii) $\mathrm{GW}(B) = 1/2$ *and* $p(T, B) = \mathbf{1}_B(T)$ *for* GW*-a.e.* T. *Consequently, for every* Θ*-stationary probability measure* μ *equivalent to* GW, *and in particular for* $\mu = \mathrm{GW}$, *the chain with initial distribution* μ *is not ergodic.*
(iii) *The measures* $\mathrm{GW}_1 = \mathrm{GW}(\cdot \mid B)$ *and* $\mathrm{GW}_0 = \mathrm{GW}(\cdot \mid B^c)$ *are* Θ*-stationary, and the chain with initial distribution* GW_r *is ergodic for* $r = 0, 1$. *The invariant sets of the chain with initial distribution* GW *are, up to* GW*-null sets,* $\emptyset$, B, B^c *and* $\mathcal{T}_\infty$.

In (ii), $\mu(B) \in (0, 1)$ because μ is equivalent to GW, and B is invariant for μ because $\mu \ll \mathrm{GW}$; this is the step from the first sentence to the second. Lyons and Peres work with labelled trees and test stationarity and invariance only on isomorphism-invariant events [4, Chapter 17]. Theorem 1 holds in that form too: stationarity on all Borel sets implies stationarity on these events, B is isomorphism-invariant, and ergodicity on all Borel sets implies ergodicity on these events. Part (iii) says that $\mathrm{GW} = (\mathrm{GW}_0 + \mathrm{GW}_1)/2$ is the ergodic decomposition of the stationary chain into two mutually singular stationary measures.

The set B is the event that the root has colour 1 in a colouring of the vertices that is a function of the tree: a vertex with two children takes the AND of the colours of its children and a vertex with three children the OR (Lemma 2). Under both rules a vertex has at least one child of its own colour, and the flow rule spreads the flow evenly over the children of the root that have the colour of the root. Under GW the colour of the root is Bernoulli(1/2), and 1/2 is the solution in $(0, 1)$ of the corresponding equation $x = \frac{3}{5}x^2 + \frac{2}{5}(1 - (1 - x)^3)$, whose other solutions are 0 and 1. The main point is that the colour is determined by the tree, without extra randomness. Section 2 proves this by a martingale and an exact second-moment recursion, Section 3 constructs the flow rule and proves Theorem 1, and Section 4 places the colouring in the framework of tree automata and recursive distributional equations.

2. The colouring

We use the Ulam–Harris labelling: a tree is a set t of finite words over $\{1, 2, \dots\}$ that contains the empty word $\varnothing$ (the root), contains every prefix of each of its words, and contains $u1, \dots, u(i-1)$ whenever it contains ui; the children of u are the words $ui \in t$, their number

$k_u(t)$ is finite, $|u|$ is the length of u, and $t^u = \{w : uw \in t\}$. For $n \ge 0$ let $t|n = \{u \in t : |u| \le n\}$. A rooted isomorphism of t onto t' maps the children of each vertex onto the children of its image. Under GW, conditionally on $k_\varnothing(T) = k$ the trees $T^1, \dots, T^k$ are independent with law GW, and for every word u, conditionally on $u \in T$, the tree T^u has law GW.

Let $\mathcal{T}_{23}$ be the closed set of trees in which every vertex has two or three children; it carries GW. Put

$$f_2(a,b) = ab, \qquad f_3(a,b,c) = 1 - (1-a)(1-b)(1-c),$$

the probability that the AND of two, or the OR of three, independent Bernoulli variables with the given parameters equals 1. On $\{0,1\}^k$, f_2 is the AND and f_3 the OR. Let $t \in \mathcal{T}_{23}$ and $n \ge 0$. Colour the vertices of generation n of t independently, each with colour 1 with probability $1/2$, and colour the vertices of generations $n-1, \dots, 0$ in turn, a vertex u with colour $f_{k_u(t)}$ of the colours of its children. Let $q_n(t)$ be the probability that the root gets colour 1.

Lemma 1. (a) *For $t \in \mathcal{T}_{23}$ with $k_\varnothing(t) = k$ and $n \ge 0$, $q_{n+1}(t) = f_k(q_n(t^1), \dots, q_n(t^k))$; $q_0 = 1/2$. Each q_n depends only on $t|n$ and is isomorphism-invariant.*

(b) *Let T have law* GW *and let $\mathcal{F}_n$ be the σ-field generated by $T|n$. Then $(q_n(T))_{n\ge0}$ is an $(\mathcal{F}_n)$-martingale, and $\mathbb{E}q_n(T) = 1/2$.*

(c) *The numbers $d_n = \mathbb{E}[2q_n(T)(1-q_n(T))]$ satisfy $d_0 = 1/2$ and*

$$d_{n+1} = \frac{d_n(3-d_n)^2}{10} = \frac{9}{10}d_n - \frac{3}{5}d_n^2 + \frac{1}{10}d_n^3, \tag{2.1}$$

and $0 \le d_n \le \frac{1}{2}\left(\frac{9}{10}\right)^n$ for all n.

Proof. (a) Generation $n+1$ of t is the disjoint union of the generations n of $t^1, \dots, t^k$. The colours of the children of the root are therefore independent, and the ith child has colour 1 with probability $q_n(t^i)$; this gives the recursion, and the remaining assertions follow from it by induction, since f_2 and f_3 are symmetric.

(b) Let V_n be generation n of T, and for $x \in [0,1]^{V_n}$ let $\Phi(x)$ be the probability that the root gets colour 1 when the vertices $v \in V_n$ receive independent colours, colour 1 with probability x_v, and the colours are propagated to the root as above. Then Φ is a polynomial, affine in each variable, determined by $T|n$. If generation $n+1$ is coloured as in the definition of q_{n+1}, a vertex $v \in V_n$ receives colour 1 with probability $g_v = f_{k_v(T)}(\frac12, \dots, \frac12)$, which is $1/4$ if v has two children and $7/8$ if it has three, independently over v. Hence $q_{n+1}(T) = \Phi\big((g_v)_{v\in V_n}\big)$. Given $\mathcal{F}_n$, the numbers $k_v(T)$, $v \in V_n$, are independent with law (p_2, p_3), so the g_v are independent with mean $\frac35 \cdot \frac14 + \frac25 \cdot \frac78 = \frac12$. Since Φ is affine in each variable, $\mathbb{E}[q_{n+1}(T) \mid \mathcal{F}_n] = \Phi(\frac12, \dots, \frac12) = q_n(T)$.

(c) Given T, colour generation n twice, independently, and let R_n and R'_n be the two resulting colours of the root. Then $\mathbb{P}(R_n \ne R'_n \mid T) = 2q_n(T)(1-q_n(T))$, so $d_n = \mathbb{P}(R_n \ne R'_n)$, and $d_0 = 1/2$. By (b), $\mathbb{P}(R_n = 1) = \mathbb{P}(R'_n = 1) = 1/2$, so the law of (R_n, R'_n) is

$$\mathbb{P}(1,0) = \mathbb{P}(0,1) = \frac{d_n}{2}, \qquad \mathbb{P}(1,1) = \mathbb{P}(0,0) = \frac{1-d_n}{2}.$$

Given $k_\varnothing(T) = k$, the trees $T^1, \dots, T^k$ are independent with law GW and the two colourings of generation $n+1$ of T restrict to two independent colourings of generation n of each T^i, independent over i. So the pairs of colours of the children of the root are independent copies of (R_n, R'_n), and (R_{n+1}, R'_{n+1}) is obtained by applying f_k to each coordinate. If $k = 2$, the AND of both coordinates is 1 with probability $((1-d_n)/2)^2$ and the AND of one coordinate with probability $1/4$, so

$$\mathbb{P}(R_{n+1} \ne R'_{n+1} \mid k = 2) = 2\Big(\frac14 - \frac{(1-d_n)^2}{4}\Big) = d_n - \frac{d_n^2}{2}.$$

If $k = 3$, the OR of both coordinates is 0 with probability $((1 - d_n)/2)^3$ and the OR of one coordinate with probability $1/8$, so

$$\mathbb{P}(R_{n+1} \neq R'_{n+1} \mid k = 3) = 2\Big(\frac{1}{8} - \frac{(1 - d_n)^3}{8}\Big) = \frac{3d_n - 3d_n^2 + d_n^3}{4}.$$

Averaging with weights $3/5$ and $2/5$ gives (2.1). For $0 \le d \le 1$,

$$0 \le \frac{d(3 - d)^2}{10} = \frac{9}{10}d - \frac{d^2(6 - d)}{10} \le \frac{9}{10}d,$$

and the bound follows by induction from $d_0 = 1/2$. □

Let E be the set of $t \in \mathcal{T}_{23}$ for which $\lim_n q_n(t)$ exists and lies in $\{0, 1\}$, and let G be the set of $t \in \mathcal{T}_{23}$ such that $t^u \in E$ for every $u \in t$. For $t \in G$ put $C(t) = \lim_n q_n(t)$.

Lemma 2. (a) *E and G are isomorphism-invariant Borel sets, C is an isomorphism-invariant Borel function on G, and $t^u \in G$ whenever $t \in G$ and $u \in t$.*

(b) $\mathrm{GW}(G) = 1$ *and* $\mathrm{GW}(\{t \in G : C(t) = 1\}) = 1/2$.

(c) *For every $t \in G$ with $k_\varnothing(t) = k$, $C(t) = f_k\big(C(t^1), \dots, C(t^k)\big)$: the colour of the root is the AND of the colours of its children if $k = 2$ and their OR if $k = 3$.*

Proof. (a) By Lemma 1(a), each q_n is locally constant, hence continuous, on $\mathcal{T}_{23}$, so E is Borel and C is Borel on G. For each word u the set $\{t : u \in t\}$ is open and closed and $t \mapsto t^u$ is continuous on it, so G, the intersection over all words u of the sets $\{t \in \mathcal{T}_{23} : u \notin t \text{ or } t^u \in E\}$, is Borel. Isomorphism invariance follows from that of the q_n, since a rooted isomorphism maps each subtree t^u onto a tree isomorphic to it. Finally $(t^u)^w = t^{uw}$.

(b) By Lemma 1(b), $q_n(T)$ converges GW-a.s., and by Lemma 1(c), $\mathbb{E}\sum_n q_n(T)(1 - q_n(T)) = \sum_n d_n/2 \le 5/2$, so $q_n(T)(1-q_n(T)) \to 0$ a.s. Hence the limit lies in $\{0, 1\}$ a.s., that is, $\mathrm{GW}(E) = 1$. For each word u, $\mathrm{GW}(u \in T,\ T^u \notin E) = \mathrm{GW}(u \in T)\mathrm{GW}(E^c) = 0$, and since there are countably many words, $\mathrm{GW}(G) = 1$. By bounded convergence, $\mathrm{GW}(C = 1) = \lim_n \mathbb{E}q_n(T) = 1/2$.

(c) Let $n \to \infty$ in Lemma 1(a); $t^1, \dots, t^k \in G$ by (a), and f_k is continuous. □

3. The flow rule

For $t \in G$ with $k_\varnothing(t) = k$, let $S(t) = \{i \le k : C(t^i) = C(t)\}$. This set is not empty, by Lemma 2(c): if $k = 2$ and $C(t) = 1$, both children have colour 1; if $k = 2$ and $C(t) = 0$, some child has colour 0; if $k = 3$ and $C(t) = 1$, some child has colour 1; and if $k = 3$ and $C(t) = 0$, all three children have colour 0. For $t \in \mathcal{T}_\infty \setminus G$, let $S(t)$ be the set of children i of the root for which t^i is infinite, which is not empty since the root has finitely many children. In both cases t^i is infinite for $i \in S(t)$, because the trees in $\mathcal{T}_{23}$ are infinite. Put $\sigma_t(i) = 1/|S(t)|$ for $i \in S(t)$ and $\sigma_t(i) = 0$ for the other children of the root, and define Θ_t for $t \in \mathcal{T}_\infty$ by $\Theta_t(\varnothing) = 1$ and, for $u \in t$ and $ui \in t$,

$$\Theta_t(ui) = \Theta_t(u)\,\sigma_{t^u}(i) \quad \text{if } \Theta_t(u) > 0, \qquad \Theta_t(ui) = 0 \quad \text{otherwise.}$$

By induction on $|u|$, $\Theta_t(u) > 0$ implies that t^u is infinite, so σ_{t^u} is defined, and then $\sum_i \Theta_t(ui) = \Theta_t(u)$. Thus Θ_t is a flow on t. For $|x| = 1$ with $\Theta_t(x) > 0$, induction on $|w|$ gives $\Theta_t(xw) = \Theta_t(x)\Theta_{t^x}(w)$ for $w \in t^x$, because $(t^x)^w = t^{xw}$; this is the consistency condition. A rooted isomorphism φ of t onto t' maps the children of the root onto the children of the root and induces, for each i, a rooted isomorphism of t^i onto $t'^{\varphi(i)}$; since G, C and finiteness are isomorphism-invariant, φ maps $S(t)$ onto $S(t')$, so $\sigma_{t'}(\varphi(i)) = \sigma_t(i)$, and by induction $\Theta_{t'}(\varphi(u)) = \Theta_t(u)$ for all $u \in t$. For each u the map $t \mapsto \Theta_t(u)$ is Borel, by Lemma 2(a). So Θ is a flow rule, and its transition kernel is

$$p(t, \cdot) = \frac{1}{|S(t)|} \sum_{i \in S(t)} \delta_{t^i}. \tag{3.1}$$

Let $B = \{t \in G : C(t) = 1\}$ and $G_0 = \{t \in G : C(t) = 0\}$, and for $r \in \{0,1\}$ let GW_r be GW conditioned on $\{t \in G : C(t) = r\}$. By Lemma 2(b), $\mathrm{GW}(B) = \mathrm{GW}(G_0) = 1/2$, $\mathrm{GW}_1 = \mathrm{GW}(\cdot \mid B)$, $\mathrm{GW}_0 = \mathrm{GW}(\cdot \mid B^c)$ and $\mathrm{GW} = (\mathrm{GW}_0 + \mathrm{GW}_1)/2$.

Lemma 3. *Let T have law GW, let $K = k_\varnothing(T)$ and $c_i = C(T^i)$, $i \le K$. For $k \in \{2,3\}$ and $c \in \{0,1\}^k$, the event $\{K = k,\ (c_1, \dots, c_k) = c\}$ has probability $p_k 2^{-k}$, and conditionally on it $T^1, \dots, T^k$ are independent and T^i has law GW_{c_i}.*

Proof. Conditionally on $K = k$, the T^i are independent with law GW, and the event $\{C(T^i) = c_i\}$ depends on T^i alone and has probability $1/2$ by Lemma 2(b). Conditioning independent random trees on events that each concern one of them leaves them independent, with the conditioned laws. □

Proof of Theorem 1. (i) Let A be a Borel set and $r \in \{0,1\}$. Let T, K and c be as in Lemma 3. On the event $\{T \in G\}$, which has probability 1, $C(T) = f_K(c)$ by Lemma 2(c), and by (3.1) $p(T,A) = N^{-1} \sum_{i:\, c_i = C(T)} \mathbf{1}_A(T^i)$ with $N = |\{i : c_i = C(T)\}|$; both $C(T)$ and N are functions of (K, c). By Lemma 3, on an event $\{K = k,\ c = c'\}$ with $f_k(c') = r$,

$$\mathbb{E}\big[p(T,A) \mid K = k,\ c = c'\big] = \frac{1}{N} \sum_{i:\, c'_i = r} \mathrm{GW}_r(A) = \mathrm{GW}_r(A).$$

Summing over the events $\{K = k,\ c = c'\}$ with $f_k(c') = r$, whose union is $\{C(T) = r\}$ up to a null set, gives

$$\int p(t,A)\, d\mathrm{GW}_r(t) = \frac{\mathbb{E}\big[p(T,A)\mathbf{1}\{C(T) = r\}\big]}{\mathrm{GW}(C = r)} = \mathrm{GW}_r(A). \tag{3.2}$$

So GW_0 and GW_1 are Θ-stationary, and so is their average GW.

(ii) For $t \in G$, every $i \in S(t)$ has $t^i \in G$ and $C(t^i) = C(t)$, so by (3.1) $p(t,B) = \mathbf{1}_B(t)$. Since $\mathrm{GW}(G) = 1$, this holds for GW-a.e. t, and $\mathrm{GW}(B) = 1/2$. The set B is isomorphism-invariant by Lemma 2(a). If μ is a Θ-stationary probability measure equivalent to GW, then $p(t,B) = \mathbf{1}_B(t)$ for μ-a.e. t and $0 < \mu(B) < 1$, so the chain with initial distribution μ is not ergodic.

(iii) Stationarity of GW_r is (3.2). Fix r, put $G'_r = \{t \in G : C(t) = r\}$, and let A be an invariant set for the chain with initial distribution GW_r. For $t \in G'_r$ the measure $p(t,\cdot)$ is carried by G'_r, so $A' = A \cap G'_r$ satisfies $p(t,A') = p(t,A) = \mathbf{1}_A(t) = \mathbf{1}_{A'}(t)$ for GW_r-a.e. t. For such t, by (3.1), $p(t,A')$ is the proportion of the $i \in S(t)$ with $t^i \in A'$; as it equals 0 or 1, $t \in A'$ holds if and only if $t^i \in A'$ for every $i \in S(t)$, and if and only if $t^i \in A'$ for some $i \in S(t)$. Let $\alpha = \mathrm{GW}_r(A') = \mathrm{GW}_r(A)$, and condition on the event $D = \{K = 2,\ c = (r,r)\}$, on which $C(T) = f_2(r,r) = r$ and $S(T) = \{1,2\}$; it has GW_r-probability $(p_2/4)/(1/2) = 3/10$. By Lemma 3, given D the trees T^1 and T^2 are independent with law GW_r. Hence

$$\alpha^2 = \mathbb{P}(T^1 \in A',\ T^2 \in A' \mid D) = \mathbb{P}(T \in A' \mid D) = 1 - \mathbb{P}(T^1 \notin A',\ T^2 \notin A' \mid D) = 1 - (1-\alpha)^2,$$

so $2\alpha(1-\alpha) = 0$ and $\alpha \in \{0,1\}$. Thus the chain with initial distribution GW_r is ergodic. If A is invariant for the chain with initial distribution GW, it is invariant for GW_0 and GW_1, which are absolutely continuous with respect to GW, so $\mathrm{GW}_0(A), \mathrm{GW}_1(A) \in \{0,1\}$; since GW_1 and GW_0 are carried by B and G_0, which partition $\mathcal{T}_\infty$ up to a GW-null set, A coincides up to a GW-null set with $\emptyset$, B, B^c or $\mathcal{T}_\infty$. □

4. Remarks

Remark 1. The flow gives no mass to the children of the root whose colour differs from that of the root. Under GW the children of the root do not all have the same colour with probability $p_2 \cdot \frac{1}{2} + p_3 \cdot \frac{3}{4} = \frac{3}{5}$, so the positivity hypothesis of [4, Proposition 17.17] fails on a set of positive measure, as it must for any flow rule with the properties of Theorem 1.

Remark 2. The proof uses only the following properties of C: it is an isomorphism-invariant Borel function, defined on a set of full GW-measure that is closed under passage to subtrees, with values in a finite set Σ; at every vertex the colour is a fixed function A_k of the colours of the k children; a colour $\sigma = A_k(c_1, \dots, c_k)$ always occurs among the c_i; and C is not GW-a.s. constant. For such a colouring the flow rule that spreads the flow evenly over the children of the colour of the root has GW as a stationary measure, by the computation in the proof of (i), and each colour class of positive measure is an invariant set, of measure less than 1 because C is not a.s. constant. In the terminology of Johnson, Podder and Skerman [3], C is an interpretation of the tree automaton (A_k); isomorphism invariance is needed for the flow rule to be equivariant.

Remark 3. The colour of Lemma 2 is the interpretation of the AND/OR automaton that corresponds to its fixed point Bernoulli(1/2), and it is unique up to GW-null sets by [3, Proposition 1.6]. Its existence also follows from [3, Theorem 1.7], which states that for two colours and an offspring distribution with finite logarithmic moment a fixed point has an interpretation if and only if the associated pivot tree, a multitype Galton–Watson tree, is subcritical or critical. There a vertex is pivotal if changing its colour, and recolouring the vertices above it by the automaton, changes the colour of the root, so that the pivotal children of a pivotal vertex are the children whose change of colour changes the colour of that vertex; for AND and OR such a child has the colour of its parent. Given that a vertex has colour 1, it has two children of colour 1, both pivotal, with probability 3/10, and three children exactly one of which has colour 1, which is then pivotal, with probability 3/10; given colour 0, it has two children of different colours, of which the one with colour 0 is pivotal, with probability 3/5, and three children of colour 0, all pivotal, with probability 1/10. The mean matrix of the pivot tree is therefore $\operatorname{diag}(9/10, 9/10)$, and the tree is subcritical. The number 9/10 is also the slope at 0 of the recursion (2.1). In the language of Aldous and Bandyopadhyay [1], Lemma 2 says that Bernoulli(1/2) is an endogenous solution of the recursive distributional equation $X \stackrel{d}{=} f_K(X_1, \dots, X_K)$, and the proof of Lemma 1(c) is an instance of their bivariate approach: a coupling of two Bernoulli(1/2) variables is determined by the probability d that they differ, and $d \mapsto d(3-d)^2/10$ has no fixed point in $(0, 1]$. A recursion of the same kind on Galton–Watson trees, in which a position is a win if some child is a loss, is studied by Holroyd and Martin [2].

Remark 4. In the example of [4, Exercise 17.22] the invariant sets A_3 and A_4 have positive measure but do not cover a set of full GW-measure, which is why the stationary measure there is not equivalent to GW. Here the invariant sets B and B^c cover everything, and GW itself is the stationary measure.

5. Computations

The repository listed under Data availability contains a Python script, `checks.py`, with its output, and a Lean file, `Bounds.lean`, with its output. With SymPy, the script verifies the fixed point $f(1/2) = 1/2$, the disagreement polynomials for AND and OR in the proof of Lemma 1(c) by enumerating pairs of colourings, the closure of the family of pair laws under the averaged step, the recursion (2.1) and the identity $\frac{9}{10}d - \frac{d(3-d)^2}{10} = \frac{d^2(6-d)}{10}$. In exact rational arithmetic it computes the laws of $q_0, \dots, q_3$ under GW (with 1, 2, 7 and 85 atoms) and the first two moments of q_4, and checks that $\mathbb{E}q_n = 1/2$ and that $\mathbb{E}[2q_n(1-q_n)]$ equals d_n from (2.1) for $n \le 4$ ($d_1 = 5/16$, $d_2 = 1849/8192$, $d_4 \approx 0.13877$); it checks the martingale identity $\mathbb{E}[q_{n+1}(T) \mid T|n = t] = q_n(t)$ on all 1, 2 and 12 trees t of depth 0, 1 and 2 and on 40 random trees of depth 3, computes the mean matrix of the pivot tree in Remark 3, and checks that $\mathrm{GW}(C = 1) = 1/2$ and that one step of the chain started from GW leaves the law of the first two generations of the tree unchanged, the colours of the children of the root being, given the first three generations, independent with

$\mathbb{P}(C(T^i) = 1) = q_2(T^i)$. A population-dynamics simulation of the law of q_n with 400,000 samples agrees with d_n to within $3 \cdot 10^{-4}$ for $n \le 80$, and at $n = 80$ more than 99.9% of the samples lie within 10^{-6} of $\{0, 1\}$. The Lean file, checked with Lean 4 [7] (version 4.33.1) and Mathlib [8], proves the disagreement identities, the recursion (2.1), the inequality $0 \le d(3-d)^2/10 \le 9d/10$ for $0 \le d \le 1$, the bound $d_n \le \frac{1}{2}(\frac{9}{10})^n$ for every sequence that satisfies (2.1) with $d_0 = 1/2$, the summability of the bound, and the entries of the pivot mean matrix. These computations test the identities and inequalities used in the proofs; they are not part of them.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [9] and working together inside the Brain framework [11], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [10], the Math Brain repository [9] and the Brain framework repository [11].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. On 6 September 2026 an agent running in OpenAI's Codex found the AND/OR construction with offspring distribution $p_2 = 3/5$, $p_3 = 2/5$, proved that the colour is a function of the tree by a martingale and second-moment argument, and proved that GW is stationary and that the colour classes are invariant. On 18 September 2026 two runs of an OpenAI model (GPT-5.6 Sol) re-derived the example, one of them proposing a two-parameter family of examples of the same kind, and on 10 October 2026 another Codex agent wrote out a proof with computer checks. On 13 October 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, wrote the proof in its present form, including the measurability of the colour and of the flow rule and the ergodicity of the two colour classes in part (iii) of Theorem 1, searched the literature, and wrote the remarks, the computer checks, the Lean file and the manuscript. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code and the Lean file are available at `https://github.com/yspennstate/nonergodic-galton-watson-flow-rule`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [9].

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